Defining and quantifying motion

STUDENT INFORMATION AND EXAMPLES

Position, distance and displacement

Choose an origin ($x = 0$) and a positive direction along a straight road. Position x (m) tells us where the car is. Distance travelled is the total length of its journey. Displacement is the change in position: $\Delta x = x_f - x_i$.

- Example: from $x = -4$ m to $x = +10$ m: $\Delta x = +14$ m. The car may have travelled farther if it turned around.
- Distance is a scalar; displacement is a vector. In 1D, + and − show opposite directions.

Rates of motion

Quantity	How to calculate	SI unit
Average speed	total distance ÷ elapsed time	m/s
Average velocity, v_{avg}	displacement ÷ elapsed time	m/s
Average acceleration, a_{avg}	(final velocity − initial velocity) ÷ elapsed time	m/s^2

Speed has no direction. Velocity includes direction. Acceleration measures how quickly velocity changes. Negative acceleration is not always slowing down: it depends on the direction of velocity.

Instantaneous speed or velocity describes one moment. An average describes the entire selected time interval.

Uniform acceleration: helpful relationships

Only when acceleration is constant: $v_f = v_i + a\Delta t$; $v_{avg} = (v_i + v_f)/2$; $\Delta x = v_{avg}\,\Delta t$. Combined: $\Delta x = v_i\Delta t + \frac{1}{2}a(\Delta t)^2$. An additional equivalent form is $v_f^2 = v_i^2 + 2a\Delta x$.

For this lesson, the key first steps are calculating average speed, average velocity and acceleration. The additional constant-acceleration equations are available for multi-step problems.

Always show: (1) equation, (2) rearrangement if needed, (3) substitution with units, (4) final value and unit. State a direction when a vector is negative or positive.

Six straight-line car journeys — data for practice

Journey	What happens	Provided measurements
1	Constant forward velocity	distance = 30 m; time = 6 s
2	Constant backward velocity	$x_i = +12$ m; $x_f = -8$ m; time = 5 s
3	Starts from rest; accelerates uniformly	$v_i = 0$; $v_f = +8$ m/s; time = 4 s
4	Brakes uniformly to rest	$v_i = +8$ m/s; $v_f = 0$; time = 4 s
5	Slows without stopping	$v_i = +10$ m/s; $v_f = +6$ m/s; time = 4 s
6	Moves forward, stops, then reverses	$v_i = +4$ m/s; $a = -2$ m/s^2; time = 4 s

Assume the stated acceleration is constant when initial and final velocities are supplied for Journeys 3–6. The observations are one-dimensional.

Worked examples — show the four stages

Journey 1 — average speed

1. Equation: average speed = total distance ÷ total time.
2. Rearrangement: Already arranged for average speed.
3. Substitution: average speed = 30 m ÷ 6 s.
4. Answer: average speed = 5 m/s.

Journey 3 — accelerating from rest

Acceleration

1. Equation: $a = (v_f - v_i) / \Delta t$
2. Rearrangement: Already arranged for a.
3. Substitution: a = (+8 m/s − 0 m/s) / 4 s
4. Answer: a = +2 m/s^2.

Average velocity (constant acceleration)

1. Equation: $v_{avg} = (v_i + v_f) / 2$
2. Rearrangement: Already arranged for v_{avg}.
3. Substitution: v_{avg} = (0 m/s + 8 m/s) / 2
4. Answer: v_{avg} = +4 m/s.

Displacement

1. Equation: $\Delta x = v_{avg} \Delta t$
2. Rearrangement: Already arranged for Δx.
3. Substitution: Δx = (+4 m/s)(4 s)
4. Answer: Δx = +16 m.

Journey 4 — braking to rest

1. Equation: $a = (v_f - v_i) / \Delta t$

2. Rearrangement: Already arranged for a.
3. Substitution: a = (0 m/s − 8 m/s) / 4 s
4. Answer: a = −2 m/s². The displacement remains +16 m.

Journey 6 — reversing direction

Time until the car stops

1. Equation: $v_f = v_i + a\Delta t$
2. Rearrangement: $\Delta t = (v_f - v_i) / a$
3. Substitution: $\Delta t = (0 \text{ m/s} - 4 \text{ m/s}) / (-2 \text{ m/s}^2)$
4. Answer: $\Delta t = 2$ s.

Final velocity

1. Equation: $v_f = v_i + a\Delta t$
2. Rearrangement: Already arranged for v_f.
3. Substitution: $v_f = 4 \text{ m/s} + (-2 \text{ m/s}^2)(4 \text{ s})$
4. Answer: $v_f = -4$ m/s.

Displacement

1. Equation: $\Delta x = v_i \Delta t + \frac{1}{2}a(\Delta t)^2$
2. Rearrangement: Already arranged for Δx.
3. Substitution: $\Delta x = (4 \text{ m/s})(4 \text{ s}) + \frac{1}{2}(-2 \text{ m/s}^2)(4 \text{ s})^2$
4. Answer: $\Delta x = 0$ m.

Total distance travelled

1. Equation: total distance = distance forward + distance back
2. Rearrangement: Separate the journey where velocity changes sign.
3. Substitution: total distance = 4 m + 4 m
4. Answer: total distance = 8 m.